

Soru - 1

How many address lines and input – output data lines are needed in each case, also give the capacity of ram.

- a) 16K x 64
- b) 1M x 32
- c) 128M x 16
- d) 4G x 8

Soru -2

An integrated circuit RAM has capacity 128K x 4,

- a) How many address and data lines
- b) How many chips are needed for 512Kx16
- c) Draw a block diagram

Soru - 3

Design 4 bit priority encoder with both PLA/PAL.
Minimize number of product terms, mark fuses.

Soru - 4

- Design four bit arithmetic circuit with two selection variables $S_1 S_0$ that generates the following arithmetic operations. Draw the logic diagram of two typical stage.

S_1	S_0	$C_{in} = 0$	$C_{in} = 1$
0	0	$G = A + B$	$G = A + B + 1$
0	1	$G = A$	$G = A + 1$
1	0	$G = A - 1$	$G = A + B' + 1$
1	1	$G = B - 1$	$G = B + 1$

Soru - 5

A 3 bit counter determines its next state by following rule:

Next state count = (current state count + 3) mod 5

Where mod is remainder operator.

Initial state is 010 use $C = C_2C_1C_0$ C_2 is most significant.

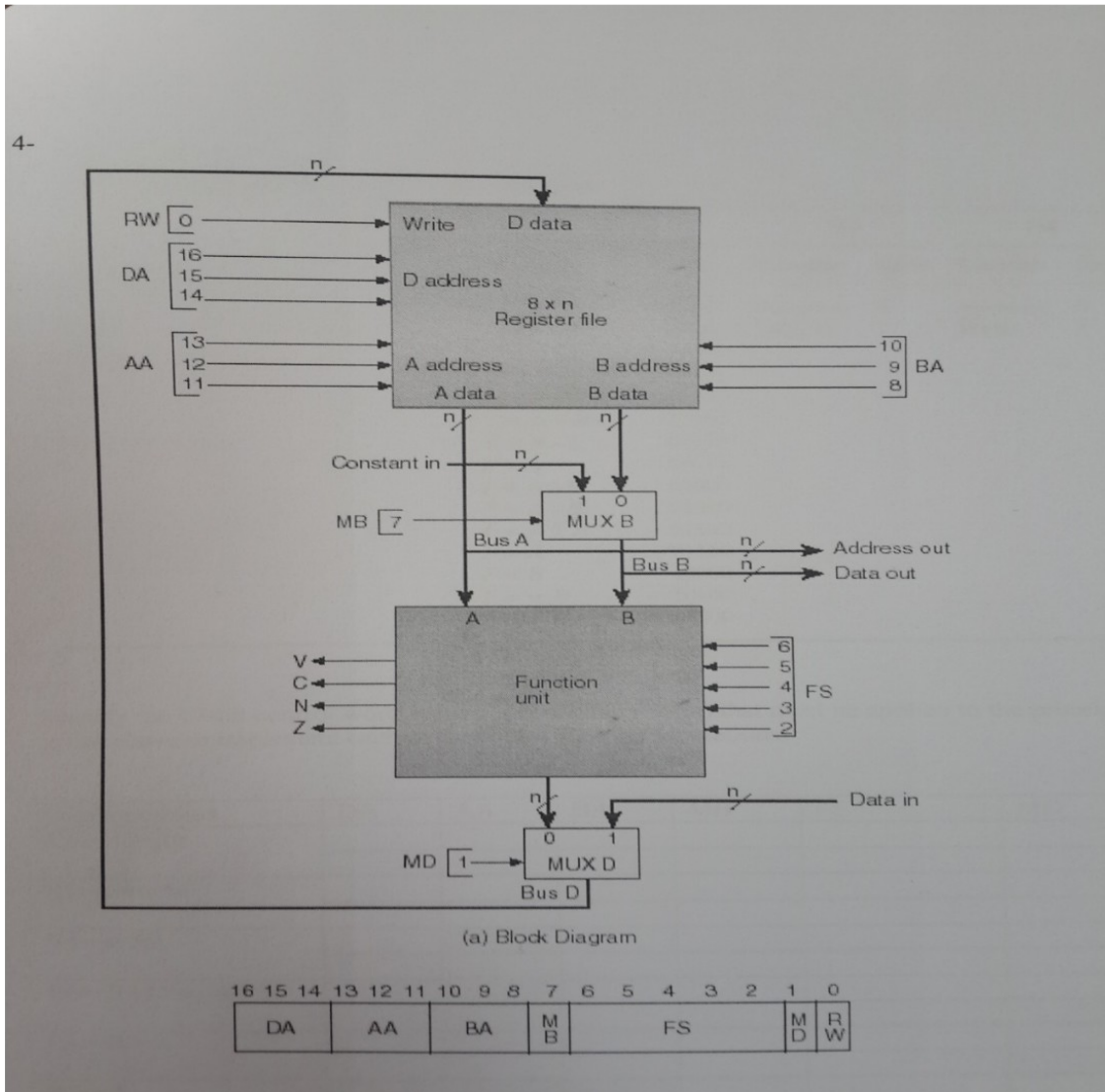
- a) Draw state diagram
- b) Implement the above counter using circuit of minimum size using 3 T flip flops. Don't care unused states, do not draw circuit
- c) Analyze self correcting property

Soru – 6

Assume that the data path described in Figure 1 uses unsigned magnitude representation and has 8 4-bit registers. Initially each register contains value of its number. The following sequence of 17 bit control words that presented datapath.

- a) For each, identify the micro-operation that is executed,
- b) Show the new contents of each destination register through the sequential execution of each micro-operation

Soru – 6 Devam



DA, AA, BA		MB		FS		MD		RW	
Function	Code	Function	Code	Function	Code	Function	Code	Function	Code
R0	000	Register	0	$F = A$	00000	Function	0	No write	0
R1	001	Constant	1	$F = A + 1$	00001	Data In	1	Write	1
R2	010			$F = A + B$	00010				
R3	011			$F = A + B + 1$	00011				
R4	100			$F = A + \overline{B}$	00100				
R5	101			$F = A + \overline{B} + 1$	00101				
R6	110			$F = A - 1$	00110				
R7	111			$F = A$	00111				
				$F = A \wedge B$	01000				
				$F = A \vee B$	01010				
				$F = A \oplus B$	01100				
				$F = \overline{A}$	01110				
				$F = B$	10000				
				$F = \text{sr } B$	10100				
				$F = \text{sl } B$	11000				

Soru - 7

- Aşağıda pseudo kodu verilen algoritmayı gerçekleyebilen en küçük veri yolunu tasarlayınız? Devrede bulunan kontrol bitlerini verilen tablodaki sütunlara yazarak her bir işlem için alacağı değerleri yazınız?
- Tasarımda bir adet toplayıcı/çıkarıcı ve yeteri kadar çoğullayıcı (mux), s...

```
X = 0
Y = 4
input Z
while (Z != 0) {
    X = X + Y
    Z = Z - 1 }
```

ini kullanınız.

Soru - 7 Çözüm

